

5-3 Motor Failure and Analysis

Motor Failure	Cause	Inspection Method	Corrective Actions
Motor cannot startup without load	Short-circuit of the coil	1. Verify if the wiring of U, V, W inside the motor is correct	If the wiring is incorrect, re-wire the U, V, W with correct wiring method.
		2. Visual inspection: Check if there is anything burnt or any surface damage inside the motor junction box and the end of the motor coil	If the coil is damaged or short-circuited, contact the manufacturer to replace the motor.
		3. Use a micro-resistance meter to measure the line resistance between any two-phase coils. If the resistance is too low, it means the coil is short-circuited	
	Coil break	1. Visual inspection: Check the coil joints for any damage such as desoldering and locate the damage point 2. Use a micro-resistance meter to measure the line resistance between any two-phase coils. If the resistance is infinite, it means the coil is broken.	If the coil is checked broken, contact the manufacturer to replace the motor.
	The coil is damp, which causes aging of the insulation and lose its function, resulting in coil damage	Use a megger of 500 V _{DC} to check whether the cold insulation resistance between the coil and the motor housing is $\geq 100\text{M}\Omega$	If the coil insulation is checked damaged, contact the manufacturer to replace the motor.
	Parameter of the motor and the AC motor drive does not match	Perform autotune again, or reset the motor parameters and check whether the motor runs smoothly	1. Adjust the AC motor drive parameters 2. Contact the manufacturer
Excessive vibration and noise after the motor starts running	Mechanical shaft stock	1. Turn the shaft to check if it rotates smoothly and check for any mechanical jam. 2. Check if the motor installation centering is accurate	1. Adjust the motor installation 2. Contact the manufacturer
	The motor centering is not accurate	1. Check if the motor installation centering is accurate 2. Check whether the straightness and concentricity of the motor mounting surface meets the specification	1. Adjust the motor installation 2. Contact the manufacturer
	Parameter of the motor and the AC motor drive does not match	Perform autotune again, or reset the motor parameters and check whether the motor runs smoothly	1. Adjust the AC motor drive parameters 2. Contact the manufacturer